

Bridging Matrix Profiles and Empirical Dynamic Modelling in Predictions for Environmental Data

Nordstat, Helsinki

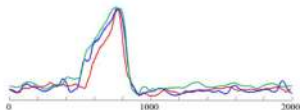
Albertas Dvirnas

2026/06/01

Different worlds to bridge

The UCR Matrix Profile Page

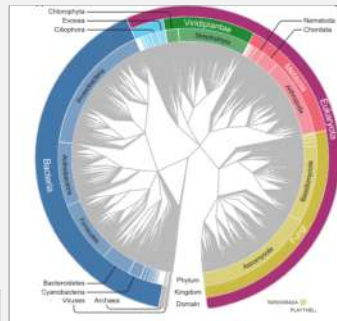
Funded by NSF IIS 1161997 II, IIS 1510741, CNS 1544969



Nonlinear forecasting as a way of distinguishing chaos from measurement error in time series

George Sugihara* & Robert M. May†

* Scripps Institution of Oceanography, University of California, San Diego, La Jolla, California 92093, USA
† Department of Zoology, Oxford University, Oxford, OX1 3PS, UK



Article [Open access](#) Published: 18 December 2025

Airborne eDNA captures three decades of ecosystem biodiversity

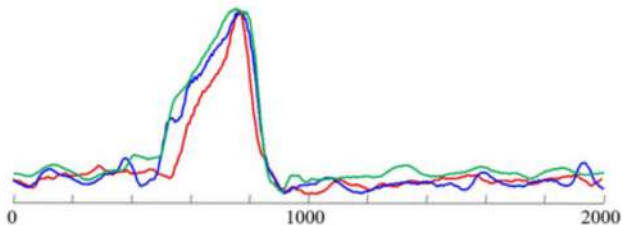
Alexis R. Sullivan, Edwin Karlsson, Daniel Srennson, Björn Brindefalk, Jose Antonio Villegas, Amanda Mikko, Daniel Betsiers, Abu Bakar Siddique, Anna-Mia Johansson, Håkan Grahn, David Sundell, Anita Norman, Per-Anders Eideen, Andreas Sjödin, Navinder J. Singh, Tomas Brodus, Meta Forsman & Per Stenberg

Nature Communications 16, Article number: 11261 (2025) | [Cite this article](#)

"Matrix Profile: what patterns repeat?" vs. "EDM: why do they repeat and what comes next?"

The UCR Matrix Profile Page

Funded by NSF IIS 1161997 II, IIS 1510741, CNS 1544969



Setup

Let $\mathbf{T} = (T_1, \dots, T_n)$ be a real-valued time series. Fix a subsequence length $m \ll n$ and define the sliding-window set

$$\mathcal{S} = \{Q_i = (T_i, \dots, T_{i+m-1}) : i = 1, \dots, n-m+1\}.$$

For each pair $Q_i, Q_j \in \mathcal{S}$, define the **z-normalised Euclidean distance**

$$d_z(Q_i, Q_j) = \|\hat{Q}_i - \hat{Q}_j\|_2, \quad \hat{Q}_k = \frac{Q_k - \mu_{Q_k}}{\sigma_{Q_k}}.$$

- Normalisation removes offset and amplitude differences. Only **shape** matters.
- Distance is bounded: $d_z \in [0, 2\sqrt{m}]$.

Matrix profile: definition

The Matrix Profile vector P

$$P_i = \min_{j: |i-j| > \lfloor m/4 \rfloor} d_z(Q_i, Q_j), \quad I_i = \arg \min_j d_z(Q_i, Q_j).$$

The exclusion zone $|i - j| > \lfloor m/4 \rfloor$ prevents **trivial self-matches**.

Link to Pearson correlation

$$d_z(Q_i, Q_j)^2 = 2m(1 - \widehat{\text{corr}}(Q_i, Q_j))$$

Minimising distance $\equiv$ maximising empirical correlation.

- **Motif:** indices where P_i is small $\rightarrow$ repeated patterns.
- **Discord:** indices where P_i is large $\rightarrow$ anomalous subsequences.

Matrix profile: statistical view

Nearest-neighbour distance over subsequence space

$\mathbf{P}$ is a nonparametric nearest-neighbour distance function over the empirical measure on $\mathbb{R}^m / \sim$, where $\sim$ is the affine-scaling equivalence class.

Under additive noise $T_t = S_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma_N^2)$, signal variance σ_S^2 (De Paepe D, Avendano DN, Van Hoecke S. Implications of z-normalization in the matrix profile. PRAM 2019):

$$\mathbb{E}[d_z(Q_i, Q_j)^2] = (2m + 2) \frac{\sigma_N^2}{\sigma_S^2 + \sigma_N^2}.$$

- Approaches 0 as noise vanishes.
- Approaches $2m + 2$ as signal drowns in noise.
- Profile value is a **signal-to-noise ratio functional**.

Nonlinear forecasting as a way of distinguishing chaos from measurement error in time series

George Sugihara* & Robert M. May†

* Scripps Institution of Oceanography, University of California, San Diego, La Jolla, California 92093, USA

† Department of Zoology, Oxford University, Oxford, OX1 3PS, UK

Empirical Dynamic Modeling (EDM)

The Equation-Free Approach

Instead of assuming parametric equations (e.g., ARIMA, Lotka-Volterra), EDM infers nonlinear ecosystem dynamics directly from observational data.

State-Space Reconstruction

By Takens' Theorem, a 1D time series x_t can reconstruct the hidden multivariate system attractor using delay coordinates:

$$\mathbf{x}_t = \langle x_t, x_{t-\tau}, x_{t-2\tau}, \dots, x_{t-(E-1)\tau} \rangle$$

- E : Embedding dimension (system complexity).
- τ : Time delay.

The reconstructed manifold preserves the topology of the true dynamic system.

EDM Forecasting: Simplex Projection

Core Assumption

Determinism: States that are close in the embedded state-space should have similar future trajectories.

Algorithm

- 1 Identify current state $\mathbf{x}_t$.
- 2 Find k nearest historical neighbors in the manifold $(N_1, \dots, N_k)$.
- 3 Track their historical forward evolution $(N_i \rightarrow N_{i+h})$.

Prediction

Exponentially weighted average of neighbors' futures (weighted by distance d_i):

$$\hat{x}_{t+h} = \sum_{i=1}^k w_i \cdot N_{i+h}$$

(Ref: our model uses $k = 20$ neighbors within a rolling $w = 65$ window).

Chronos2: A Zero-Shot Forecasting Baseline

What is Chronos2?

- A large language model (T5 architecture) pretrained on a large corpus of real and synthetic time series data.
- Produces **probabilistic forecasts** via quantile regression - no fine-tuning on the target series required (zero-shot).
- Treats forecasting as a **language modeling task**: time series values are tokenized into discrete bins.

Role in This Work

Used as a **state-of-the-art black-box baseline** to benchmark EDM. Unlike EDM, Chronos2 requires no embedding dimension selection, no state-space reconstruction, and has no mechanistic interpretation - making the comparison methodologically meaningful.

The SweBITS Kiruna Airborne eDNA Dataset

Data Generation & Structure

- **Source:** Archived glass-fiber air filters from a Kiruna monitoring station (Arctic boreal zone).
- **Taxonomic Scope:** Weekly relative abundance for $N > 2700$ organism genera.
- **Time Span:** 34 years spanning from 1974 to 2008.
- **Format:** High-dimensional multivariate time series.

Systematic Missingness

- **Missing Years:** Only *even-numbered years* were sequenced. Induces alternating 52-week data blocks and 52-week gaps.
- **Missing Weeks:** Filter analysis restricted to weeks with mean temperature $> 0^{\circ}\text{C}$.

Statistical Challenge

Models must handle deterministic, structured missing data.

The matrix profile window size w and the EDM embedding dimension E are the same object in disguise.

Both define the length of the delay-coordinate vector:

$$\underbrace{Q_i = (T_i, T_{i+1}, \dots, T_{i+m-1})}_{\text{Matrix Profile subsequence}} = \underbrace{\Phi_{E, \tau=1}(t_i)}_{\text{Takens delay vector with } \tau=1}, \quad (1)$$

so a matrix profile of window size w is literally computing nearest-neighbour distances in an w -dimensional delay-embedding space.

Time Series Cross-Validation

The Non-I.I.D. Challenge

Standard k -fold fails due to serial correlation. We must preserve temporal order: train on $\mathcal{Y}_{1:t}$, test on $\mathcal{Y}_{t+h}$.

Expanding Window

Fixed origin, growing history.

- **Train:** $[1, t] \rightarrow$ **Test:** $[t + 1, t + h]$
- **Train:** $[1, t + 1] \rightarrow$ **Test:** $[t + 2, \dots]$
- **Pro:** Maximizes data for long-memory models.

Rolling Window

Moving origin, fixed size w .

- **Train:** $[t - w, t] \rightarrow$ **Test:** $[t + 1, t + h]$
- **Train:** $[t - w + 1, t + 1] \rightarrow \dots$
- **Pro:** Adapts to non-stationarity.

The Horizon Problem

Strict calendar forecasts ($t + h$) land in missing periods (winter or odd years).

Modified Rolling Origin Design

- **Operational Time:** Horizon h shifts over *observed* indices only, ignoring unobserved calendar gaps.
- **Cross-Year Jumps:** "Next step" predictions bridge the end of year Y directly to the start of year $Y + 2$.
- **Dual Evaluation:** CV folds independently assess continuous intra-season steps vs. inter-season jumps.

Evaluation in Compositional Space

Data Transformation

eDNA relative abundances live on a simplex $\mathcal{S}^D$. We apply the **Centered Log-Ratio (CLR)** to evaluate in unconstrained space $\mathbb{R}^D$.

$$\text{CLR}(x_i) = \ln \left(\frac{x_i}{g(\mathbf{x})} \right)$$

where $g(\mathbf{x})$ is the geometric mean of the composition vector.

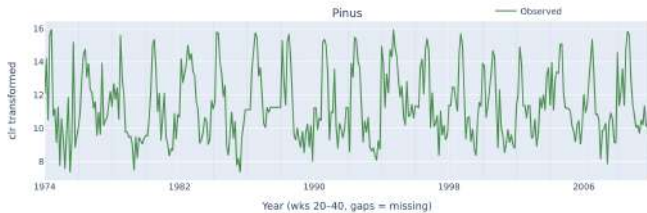
Metrics (on CLR)

- **RMSE:** Penalizes large errors.
- **MAE:** Robust to outliers.
- **Pearson ρ :** Captures dynamic shape and structural similarity.

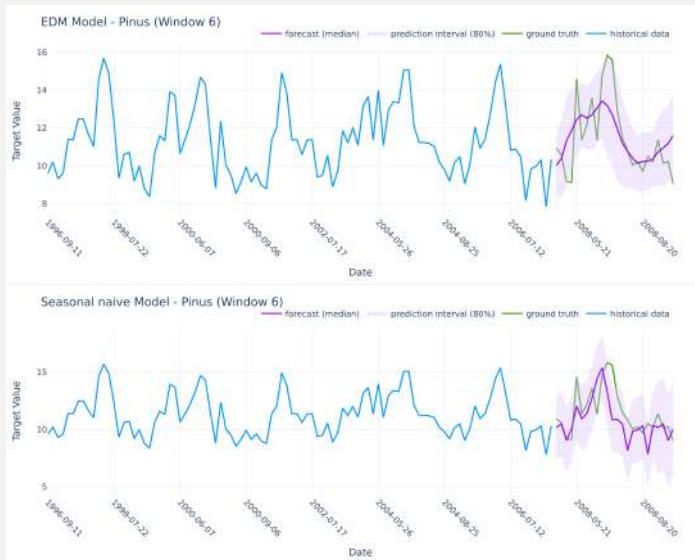
Baselines

- **Naive:** $\hat{y}_{t+h} = y_t$
- **Seasonal Naive:**
 $\hat{y}_{t+h} = y_{t+h-s}$ (copies previous observed season).

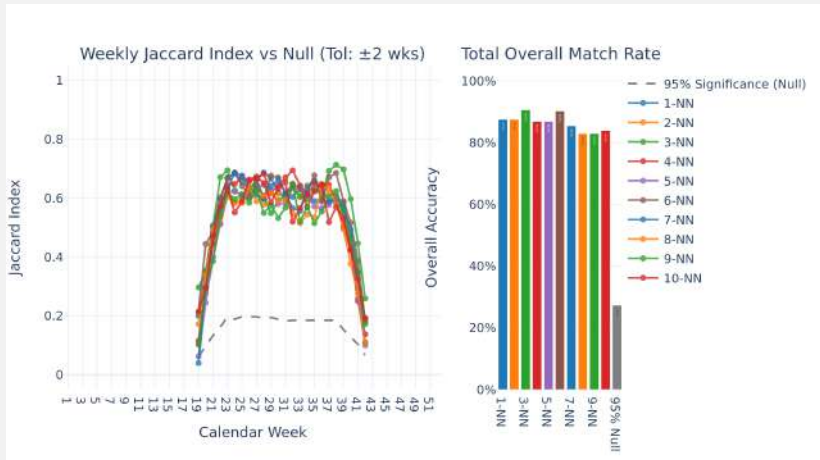
Example



Results: visual comparison of forecasting example



Jaccard summary for an example time series

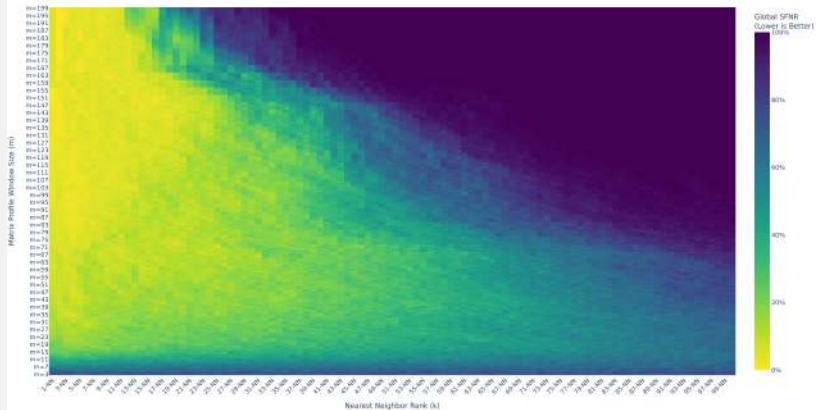


Performance dependence on window width



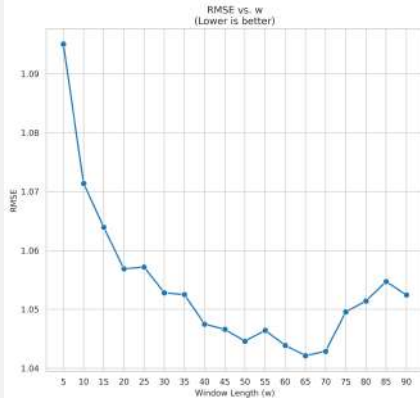
Heatmap

Seasonal False-Neighbor Rate by Window Size and Rank (ToI: ± 2 wks)



Performance metrics

EDM performance by window length (w)



Results: method benchmarking

Table: Performance comparison of forecasting models. Best results for each metric are highlighted in bold.

Model	RMSE	MAE	Pearson ρ
Seasonal naive	1.3876	1.0413	0.7612
Naive	1.3287	0.9732	0.7706
EDM $w=65,k=20$	1.0468	0.7759	0.8538
Chronos2	1.0546	0.7721	0.8511

Results: method benchmarking

Table: Number of Series Won by Each Model (Total Series Evaluated: 1494)

Model	RMSE	MAE	Pearson ρ	RMSE Win %
EDM $w=65,k=20$	851	649	853	57.0%
Chronos2	623	821	170	41.7%
Naive	20	24	177	1.3%
Seasonal naive	0	0	294	0.0%

Conclusions

- **The Methodological Bridge:** Matrix Profile subsequences and EDM embeddings map to the same delay-coordinate space, linking pattern discovery with dynamical forecasting.
- **Interpretability vs. Black Box:** Foundational models (Chronos2) achieve highly competitive accuracy, but EDM provides a transparent, mechanistic alternative for ecological inference.
- **Open Statistical Frontiers:** The matrix profile currently lacks robust distributional theory, presenting rich opportunities for statisticians.
- **eDNA as a Dynamical Tool:** Decadal eDNA time series contain powerful, forecastable ecosystem signals, provided structural gaps and compositional constraints are handled correctly.

Funding and Acknowledgments



Special Thanks

- **Åke Brännström's Group**
Home of the project
- **Per Stenberg's Group**
Home of the data, Swedish biodiversity in time and space, SweBITS
- **Project Team**
Ohmuro Kouhei, Henrik Høiberg Jessen